Enterprise Infrastructure Plan

ABC Startup Solutions

ITEP 414 – System Administration and Maintenance | Week 2

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Enterprise Infrastructure Planning for a Startup Company

Part 1: Company Profile

Company Name: ABC Startup Solutions

Industry: Software Development / IT Solutions

Company Type: Newly established startup

Employees: 20

Office: Single-floor office in the Province of Laguna, Philippines

Departments: Information Technology, Human Resources, Finance, and Sales.

Company Overview

ABC Startup Solutions is a newly established software development company that requires an initial IT infrastructure to support its 20 employees. The infrastructure must provide dependable workstations, centralized services, secure networking, business software, backup capability, and room for future growth.

Department Distribution

Department	Employees	Primary Function
Information Technology	5	Software development, IT administration, infrastructure and support
Human Resources	4	Employee records and administrative activities
Finance	5	Financial records, budgeting and accounting
Sales	6	Client relations, sales and business development
Total	20	

Part 2: Enterprise Hardware Inventory

The inventory is sized for 20 employees while providing the IT team with enough portable equipment, power protection, storage, and backup capacity to keep basic operations supportable.

Asset ID	Hardware	Qty.	Department	Purpose
HW-001	Desktop Computer	20	All departments	Primary workstation for each employee.
HW-002	Business Laptop	5	IT	Administration, troubleshooting, presentations, and emergency replacement.
HW-003	Server	1	IT	Centralized services, file services, monitoring, and internal workloads.
HW-004	Business Router	1	IT	Routing, DHCP, NAT, WAN/LAN management.
HW-005	Managed Switch	1	IT	Core wired LAN connectivity and future VLAN capability.
HW-006	Network Printer	2	HR / Finance	Shared printing for administrative documents.
HW-007	UPS	4	IT / Server / Network	Power protection for critical equipment.
HW-008	Wireless Access Point	3	All departments	Business Wi-Fi coverage and controlled wireless access.
HW-009	NAS Storage	1	IT	Centralized backup target and shared storage.
HW-010	External Backup Drive	2	IT	Offline backup rotation and recovery copies.
HW-011	Monitor	20	All departments	Dedicated display for each desktop workstation.

Justification

Twenty desktops directly match the 20-employee scenario. Five laptops give the IT team mobility and temporary replacement capacity. A dedicated server centralizes selected services. The NAS provides a local backup target, while two external drives support an offline rotation. Three access points are appropriate for a single office floor, and a managed switch provides a foundation for structured expansion.

Part 3: Enterprise Software Inventory

The required software list from the activity is retained. Version fields intentionally use release descriptions rather than invented build numbers; production deployment should be pinned to the current supported release during procurement.

Software	Version / Release	License	Purpose
Windows 11 Pro	Current supported release	Commercial	Standard business endpoint operating system.
Ubuntu Server	Current LTS release	Open source	Server operating system for centralized services.
Microsoft Office	Current supported release	Commercial / subscription	Productivity, documentation, spreadsheets, and presentations.
Visual Studio Code	Current stable release	Free / open source	Development and administration editing environment.
Git	Current stable release	Open source	Version control for scripts, documentation, and software.
GitHub Desktop	Current stable release	Free	Graphical Git and repository workflow.
VirtualBox	Current supported release	Open source / license dependent	Virtual-machine testing and isolated lab environments.
Google Chrome	Current stable release	Free	Web browsing and web-based business services.
Microsoft Defender	Current supported	Included with Windows	Endpoint malware protection and security monitoring.
AnyDesk	Current supported release	Commercial / license dependent	Authorized remote support and troubleshooting.
7-Zip	Current stable release	Open source	File compression and archive extraction.

Software Governance

Only approved and appropriately licensed software should be installed. IT should maintain a software inventory, patch schedule, and standard workstation configuration so that endpoints remain consistent and supportable.

Part 4: Enterprise Network Inventory

The network inventory establishes the physical components required to connect the office to the internet and provide a secure internal LAN.

Asset ID	Equipment	Qty.	Purpose
NET-001	ISP Modem / ONT	1	Terminates the business internet service.
NET-002	Business Router	1	Routing, DHCP, NAT, and WAN/LAN management.
NET-003	Firewall	1	Security boundary and traffic-control point.
NET-004	Managed Gigabit Switch	1	Connects wired endpoints, server, printers, and APs.
NET-005	Wireless Access Point	3	Controlled wireless connectivity across the office.
NET-006	Patch Panel	1	Organizes and terminates structured Ethernet cabling.
NET-007	CAT6 Cable	2 boxes	Structured wired connections and future expansion.
NET-008	RJ45 Connectors	100 pcs	Ethernet termination and cable preparation.

Addressing and Segmentation Approach

The LAN should use private IPv4 addressing with DHCP for normal workstations and reserved or static addresses for infrastructure devices. As the company grows, VLANs should separate administrative, employee, server, and guest traffic. Firewall rules should control traffic between segments and provide logging for security review.

Part 5: Enterprise Network Diagram

The topology follows the required logical path: Internet → ISP Modem → Router → Firewall → Managed Switch, with the switch providing access to the server, printers, wireless access points, and four departments.

INTERNET

↓

Business ISP

↓

ISP MODEM / ONT

↓

ROUTER

↓

FIREWALL

↓

MANAGED SWITCH

■ ↓ ■

SERVER PRINTERS WIRELESS APs

■ ↓ ↓ ■

IT HR FINANCE SALES

Diagram Title: ABC Startup Solutions: Enterprise Network Topology
Required creation tool: Draw.io
Required exports: PNG and PDF
Student Name: Benedict James A. Gellido

The repository should contain the Draw.io-produced exports in the diagrams/ folder.

Part 6: System Administration Roles

6.1 Helpdesk Technician

Responsibilities: Receive and document incidents, perform first-line troubleshooting, prepare endpoints, install approved software, handle routine access requests, and escalate issues that require deeper infrastructure changes.

Skills: Hardware and software troubleshooting, operating-system support, customer service, ticket documentation, basic networking, communication, and problem solving.

Common Tools: Ticketing systems, remote-support tools, endpoint utilities, Windows diagnostic tools, browser tools, and basic network-testing commands.

Certification: CompTIA A+ is a relevant foundational credential for IT support and troubleshooting.

6.2 Network Administrator

Responsibilities: Configure and maintain switches, routers, wireless access points, firewalls, IP addressing, network availability, and connectivity. Monitor performance and troubleshoot outages.

Skills: TCP/IP, subnetting, routing, switching, wireless networking, VLANs, firewall concepts, monitoring, documentation, and troubleshooting.

Common Tools: Device management interfaces, packet-capture/diagnostic tools, ping, traceroute, IP utilities, monitoring dashboards, and cable testers.

Certification: CompTIA Network+ is a relevant foundational networking credential.

6.3 Linux System Administrator

Responsibilities: Install and configure Linux servers, manage users and permissions, maintain filesystems and services, monitor performance, apply updates, troubleshoot failures, and maintain secure server configurations.

Skills: Linux command line, user/group management, storage, networking, service management, logs, security, scripting, backups, and troubleshooting.

Common Tools: SSH, systemd utilities, log tools, package managers, shell scripts, storage utilities, and monitoring tools.

Certification: Linux Foundation Certified System Administrator (LFCS) aligns with practical Linux administration competencies.

6.4 Cloud Administrator

Responsibilities: Provision and monitor cloud resources, manage identities and access, configure storage and compute, maintain virtual networking, apply governance/security controls, and support backup/recovery operations.

Skills: Cloud networking, virtualization, identity and access management, storage, compute, monitoring, security, automation, and cost awareness.

Common Tools: Cloud portals, CLI tools, infrastructure-as-code tools, monitoring services, identity consoles, and backup/recovery services.

Certification: Microsoft Certified: Azure Administrator Associate (AZ-104) is a role-based administrator certification; AWS Certified Cloud Practitioner is a foundational cloud credential.

How the Four Roles Work Together

The Helpdesk Technician is normally the first point of contact for employees, records incidents, performs basic troubleshooting, and escalates infrastructure problems. The Network Administrator handles connectivity, switching, routing, wireless, and firewall issues. The Linux System Administrator manages the organization's Linux server workloads and the services that depend on them. The Cloud Administrator manages cloud-hosted resources, identities, virtual networking, storage, monitoring, and recovery processes. These roles must share documentation, change records, monitoring information, and incident details so that a problem can be traced from the user's device through the network to an on-premises or cloud service. In a small startup, the same IT employee may cover several roles, but clear responsibility boundaries still reduce confusion and improve accountability.

Situation	Primary Owner	Collaboration
User cannot access an application	Helpdesk	Escalates to Network/Linux/Cloud based on the affected service.
Office-wide connectivity failure	Network Administrator	Coordinates with Helpdesk and ISP; checks router, firewall, switch, and APs.
Server service unavailable	Linux Administrator	Works with Network and Helpdesk; checks service status, logs, storage, and connectivity.
Cloud resource unavailable	Cloud Administrator	Works with Helpdesk and service owners; checks identity, compute, networking, and monitoring.

Part 7: Infrastructure Recommendations

7.1 Internet Provider

Use a business-grade fiber internet service with a service-level agreement appropriate for a 20-person software company. Prioritize stable upload/download performance, low latency, responsive business support, and an option for a static public IP if a future service requires it. A secondary connection can be introduced later for failover.

Reason: A software company depends on repositories, cloud services, communication, remote support, software updates, and customer-facing systems, so unreliable internet can interrupt multiple departments at once.

7.2 Server Specifications

Component	Recommended Baseline	Reason
CPU	8-core server-class processor	Headroom for multiple infrastructure services.
Memory	32 GB ECC RAM	Supports server workloads and improves memory-error resilience.
Storage	2 × 1 TB SSD in RAID 1 + separate backup target	Redundancy for the primary server volume; backup remains separate.
Network	2 × 1 GbE or faster NICs	Flexibility for traffic and future expansion.
Power	UPS-backed	Reduces disruption during short power events.
OS	Ubuntu Server LTS	Long-term support and strong fit for Linux server workloads.

7.3 Backup Strategy

Use a 3-2-1-oriented strategy: maintain multiple copies of important data, use different storage media, and keep at least one copy offline or otherwise isolated from the primary environment. The NAS can be the centralized backup target, while the two external drives can be rotated for offline copies. Backups should be scheduled, monitored, encrypted where appropriate, and tested through periodic restoration exercises.

7.4 Security Recommendations

• Place the firewall at the network perimeter and deny unnecessary inbound traffic by default.

• Use separate administrative and standard user accounts and apply least privilege.

• Enable MFA for cloud and privileged accounts where supported.

• Patch operating systems, browsers, applications, firmware, and security tools.

• Use endpoint protection and centralized security logging where practical.

• Separate guest wireless access from internal business systems.

• Document changes to routers, firewalls, servers, and user access.

7.5 Antivirus / Endpoint Protection

Use Microsoft Defender on Windows endpoints as the baseline endpoint protection layer, kept current through supported update and management mechanisms. Linux servers should be hardened, patched, monitored, and protected according to the services they run.

7.6 Password Policy

• Use unique passwords for each account and prohibit shared administrator credentials.

• Prefer long passphrases and password-manager-supported credentials.

• Require MFA for privileged, cloud, and other high-value accounts where supported.

• Disable or remove accounts promptly when employees leave or no longer need access.

• Review privileged access regularly and keep administrative actions attributable to named users.

7.7 Expansion Plan

The first expansion stage should add redundant internet connectivity, additional switch capacity, spare access-point capacity, and centralized endpoint management. As the company grows, VLANs should separate departments and guests, server roles should be reviewed for virtualization or additional hosts, and cloud services can be adopted where they improve availability or reduce operational burden. Procurement should use standard models and documented configurations so expansion does not create an inconsistent environment.

Part 8: Personal Reflection

This project taught me that system administration begins with planning rather than immediately installing devices or software. A reliable infrastructure needs to be connected to the organization's actual requirements. By working with the ABC Startup Solutions scenario, I had to think about the number of employees, the departments that need technology, the equipment required to support daily work, and the network and security components that connect everything together.

The most challenging task was designing the network topology because the diagram had to include many components while still remaining logical and easy to understand. I had to consider the relationship between the internet connection, modem, router, firewall, switch, server, wireless access points, printers, and department workstations. I also found the inventory planning challenging because quantities need to be realistic. It is not enough to list equipment; each item should have a clear purpose and should support the organization's operations.

Planning is important before deployment because it reduces unnecessary purchases, prevents configuration problems, and provides a clear direction for implementation. A good infrastructure plan also helps an organization prepare for security, backups, maintenance, and future expansion. Without planning, a company may end up with incompatible equipment, weak security controls, insufficient capacity, or an infrastructure that is difficult to maintain.

This project can help me become a better System Administrator because it trained me to look at IT from an organizational perspective. Instead of thinking only about individual computers, I learned to consider the whole environment: users, hardware, software, networking, security, backups, documentation, and future growth. It also strengthened my technical communication skills because the decisions need to be documented clearly for other IT staff and management. As I continue learning, I can use this planning mindset when working with real systems and laboratory environments.

References and Source Notes

The Week 2 LSPU module is the primary source for the project scope, scenario, required sections, minimum inventory items, network-diagram requirements, documentation structure, and submission checklist. Current official documentation was used in the guide to support selected technical role and platform notes.

Source	Use in this report
LSPU ITEP 414 Week 2 Portfolio Project — Enterprise Infrastructure Planning for a Startup Company	Primary assignment requirements, scenario, deliverables, and grading criteria.
Ubuntu Server Documentation — https://ubuntu.com/server/docs/	Ubuntu Server role, LTS/support considerations, and administration context.
Linux Foundation LFCS — https://training.linuxfoundation.org/certification/LFCS/	Linux administration competencies and LFCS certification alignment.
Microsoft Learn — Azure Administrator Associate — https://learn.microsoft.com/en-us/credentials/certifications/azure-administrator/	Cloud administrator responsibilities and AZ-104 certification alignment.
Microsoft Learn — AZ-104 Study Guide — https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/az-104	Current Azure administration skill areas and study expectations.
AWS Documentation — AWS Certified Cloud Practitioner — https://docs.aws.amazon.com/aws-certification/latest/cloud-practitioner-02/cloud-practitioner-02.html	Foundational cloud certification reference.

Course Deliverable Alignment

Required Item	Included
Company Profile	Part 1
Hardware Inventory	Part 2
Software Inventory	Part 3
Network Inventory	Part 4
Network Diagram	Part 5
IT Roles Research	Part 6
Infrastructure Recommendations	Part 7
Personal Reflection	Part 8